Python — Functions & Return Values

Grade 3 | Coding

What is a Function?

A function is a reusable block of code.

Think of it like a machine:

1. You give it something (input)
2. It does a job
3. It gives something back (output)

Functions help us avoid repeating code!

Real-Life Example

Imagine a juice machine:

1. You put in oranges (input)
2. The machine squeezes them (process)
3. You get orange juice (output)

A function works the same way!
Input -> Process -> Output

The def Keyword

We use def to create a function:

```
def say_hello():
    print("Hello, friend!")
```

- def means 'define'
- say_hello is the function name
- The colon : starts the function body
- Indented code belongs to the function

Calling a Function

Creating a function doesn't run it!
You must call it by name:

```
def say_hello():
    print("Hello, friend!")

say_hello()   # This runs it!
```

Output: Hello, friend!

Try It Yourself!

Type this in Python:

```
def greet():
    print("Namaste!")
    print("Welcome to coding!")

greet()
```

What do you see on screen?

Parameters (Inputs)

Parameters let us send information
into a function:

```
def greet(name):
    print("Hello, " + name + "!")
```

- name is a parameter
- It's like a blank space the function fills in
- We give it a value when we call it

Example with a Parameter

```
def greet(name):
    print("Hello, " + name + "!")

greet("Aarav")
greet("Priya")
```

Output:

Hello, Aarav!

Hello, Priya!

Same function, different inputs!

Multiple Parameters

Functions can take more than one input:

```
def add(a, b):
    print(a + b)

add(3, 5)      # Output: 8
add(10, 20)    # Output: 30
```

Separate parameters with commas!

What is return?

return sends a value BACK from
the function:

```
def add(a, b):
    return a + b
```

- print() shows on screen
- return gives the answer back to your code
- You can save it in a variable!

Return Example

```
def add(a, b):
    return a + b

result = add(4, 6)
print(result)
```

Output: 10

The function gave back 10,
and we stored it in result!

Using Returned Values

```
def double(n):
    return n * 2

x = double(7)
print(x)         # Output: 14
print(double(3))  # Output: 6
```

You can use return values anywhere in your code!

Practice 1

Write a function called square that
takes a number and returns it squared:

```
def square(n):
    return n * n

print(square(5))   # What prints?
print(square(9))   # What prints?
```

Practice 2

Write a function called is_even that returns True if a number is even:

```
def is_even(n):
    return n % 2 == 0

print(is_even(4))   # True
print(is_even(7))   # False
```

Hint: % gives the remainder!

Key Takeaways

- def creates a function
- Call it by name with ()
- Parameters are inputs
- return sends a value back
- Functions make code reusable & neat!

Functions = your coding superpower!

Great Job!

You learned how to:

- Create functions with def
- Add parameters for inputs
- Use return to get values back

Keep practising — you are becoming a Python pro!